

$$\begin{aligned}
SKK &= (\lambda x. \lambda y. \lambda z. (x \ z) \ (y \ z)) \ (\lambda x. \lambda y. x) \ (\lambda x. \lambda y. x) = \\
&(\lambda y. \lambda z. ((\lambda x. \lambda y. x) \ z) \ (y \ z)) \ (\lambda x. \lambda y. x) = \\
&(\lambda z. ((\lambda x. \lambda y. x) \ z) \ ((\lambda x. \lambda y. x) \ z)) = \\
&(\lambda z. (\lambda y. z) \ (\lambda y. z)) = \\
&(\lambda z. (\lambda y. z) \ (\lambda y. z)) = (\lambda z. z) = I
\end{aligned}$$